

Supervisor Briefing Note — Week 1

To: Prof. Majid Shaalan **From:** Zixuan Liang **Subject:** CISC 699 Summer 2026 — proposed applied project (SafeExec: hardened Python execution sandbox for LLM agents) **Date:** (*populate before sending*)

1. One-paragraph summary

I am proposing a hardened, threat-modeled Python execution sandbox designed for use as a tool by LLM-powered software agents (agents in the lineage of Claude Code, Code Interpreter, and similar systems). The artifact is a small HTTP service that accepts Python code and runs it inside an isolated environment, with two pluggable isolation back-ends (a hardened Docker configuration and a gVisor-backed configuration). The intellectual contribution is a curated adversarial test suite — roughly 40 programs across categories like resource exhaustion, fork bombs, syscall abuse, and attempted exfiltration — and an empirical comparison of isolation strength and performance overhead across the two back-ends. The full charter draft and a three-candidate comparison are in [docs/01-launch-packet/](#).

2. Why this project for me

This project sits at the intersection of (a) production software engineering, (b) systems and Linux security primitives, and (c) the agentic-LLM domain that is currently a major area of industry hiring — which aligns with my career objective of moving toward software-engineering roles at frontier AI labs. The project does not require model training, novel ML methodology, or specialized hardware, which means I can focus my 14 weeks on doing the systems and evaluation work *well* rather than learning ML internals from scratch alongside it.

3. What I'm asking from you in W2

The launch packet asks me to "list the decisions still needed from a Week 1 conversation." Mine are below, ordered roughly by impact.

Decisions

1. **Project approval at the proposed scope.** Is the Python-only, two-back-end, ≥ 40 -program adversarial suite the right size for CISC 699? I have a fallback plan (Candidate A in [candidate-projects.md](#)) if you'd prefer a less systems-heavy artifact.
2. **Framing of the "contribution."** I am positioning the adversarial test suite and its evaluation methodology as the project's intellectual contribution, rather than positioning a novel isolation technique as the contribution. Is that framing acceptable to you?
3. **Open-source release.** Are you comfortable with me releasing the artifact (code + adversarial suite) under an open-source license at the end of the term, or would you prefer it remain private?
4. **Reference-agent demo.** I plan to build a small Claude-API-driven agent (≤ 50 lines) that uses the sandbox as a tool, included strictly as integration demo. Is that scope acceptable, or would you prefer no third-party API in the demo path?

Clarifications / preferences

5. **Repository / version-control arrangement.** Is a private GitHub repo with a final share link to the supervisor acceptable, or does the program have a preferred host (HU GitLab, etc.)?
6. **Report and presentation templates.** Are there CISC 699 templates (Word, LaTeX, Beamer) you expect me to use, or am I free to choose?
7. **Citation style.** IEEE, ACM, APA — any preference?
8. **Meeting cadence.** How often would you like to meet? My default proposal: standing 30-minute check-in every two weeks, plus dedicated longer sessions at W2 (charter approval), W4 (design review), W7 (midpoint), W11 (draft feedback).

Risks I'd like your read on

9. **Adversarial-suite quality risk.** The suite's quality is the project's central evaluation contribution. I'd like your input on what constitutes "credible" — categories, sample size per category, the right balance between hand-authored and CVE-derived programs.
10. **gVisor variability risk.** gVisor performance is reportedly variable; my evaluation plan addresses this with sample-size analysis and percentile-based reporting. I'd like your sanity-check on the proposed protocol before W4 design review.

4. What I have done already in W1

- Drafted the project charter, problem statement, three-candidate comparison, feasibility memo, and supervisor briefing (this document).
- Set up the project repository at <https://github.com/liangzixuan/cisc-699> with the structure described in [README.md](#).
- Started the engineering log ([engineering-log.md](#)) and AI-use disclosure log ([docs/ai-use-log.md](#)).
- Identified five source categories for the annotated bibliography and the gaps I still need to fill (<https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/annotated-bibliography.md>).

5. What I plan to do in W2

- incorporate supervisor feedback into the charter
- expand the bibliography toward the W11 ≥ 15 -source target
- start W3 lit-synthesis and requirements work

6. Attachments

- <https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/project-charter.md>
- <https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/problem-statement.md>
- <https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/candidate-projects.md>
- <https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/feasibility-memo.md>
- <https://github.com/liangzixuan/cisc-699/blob/main/docs/01-launch-packet/annotated-bibliography.md> (in-progress)

